

STATEMENT OF WORK

Title: Mechanical/Electrical/Plumbing (MEP) Architectural & Engineering (AE) Indefinite Delivery/Indefinite Quantity (IDIQ) Contract

I. BACKGROUND INFORMATION:

The National Institute of Standards and Technology (NIST) Gaithersburg, Maryland campus is a Research and Development (R&D) facility with 52 buildings; including a combination of administrative and laboratory spaces, located on a 578-acre campus. Other facilities are located within Montgomery County, Maryland.

Utilities are distributed to the individual buildings by underground pipe and electrical distribution systems from central plant services owned and operated by NIST. NIST campus facilities provide chilled water (42F), steam (150 psig), and compressed air (100psig) from a centralized utility generation plant located on the campus. Most, but not all buildings, are supplied from these centralized utilities. Commercial high-voltage electricity is received at a main sub-station and distributed radially (13.8kV) by multiple facility spot networks.

Most R&D buildings are technically sophisticated laboratory facilities with specialized environmental and utility systems required for research and experimental activities. The majority of the laboratories are wet-chemical laboratories. There are limited numbers of cleanrooms, radiological laboratories, and biohazard laboratories. Laboratory facilities typically utilize specialized services such as uninterrupted power, compressed gas systems, clean-room technologies, and tight-temperature and -humidity tolerance control systems for conducting research. Facilities' equipment is generally controlled by and reports to a central building automation system (BACnet and limited to pre-selected vendors). Building level equipment tends to have direct digital controls but pneumatics are still used on some terminal units.

Existing buildings are 3-60 years old and are generally steel superstructure with block and brick wall construction. Several facilities have specialized layouts with below grade concrete sub-basements and/or combination window/panelized storefront facade. Facility interior and exterior rehabilitation and renovations are continuous as buildings and systems age, or scientific projects require new layouts to accommodate on-going research. Buildings typically remain occupied during renovation activities with outages affecting utilities needing to be scheduled in advance or done during semiannual site utility outages.

II. PURPOSE:

The Government seeks to award up to three (3) offerors to provide MEP AE services via an IDIQ contract for planning, evaluation, design and engineering services for the maintenance, modification, and improvement of specialized facility mechanical and electrical systems.

III. PLACE OF PERFORMANCE:

Performance shall take place within the NIST Gaithersburg campus and other facilities located in Montgomery County, Maryland as defined by the Task order.

IV. PERIOD OF PERFORMANCE:

NIST intends to conduct a competitive acquisition that consists of one (1) base-year contract plus four (4), one-year ordering periods. Each task order will specify its period of performance.

V. TASKS:

This Statement of Work (SOW) sets forth the general requirements for the performance of various services required under this Contract. The services described or specified shall not be deemed to constitute a comprehensive list having the effect of excluding services not specifically mentioned. Additional services may be identified at the task-order level. Firms must provide multi-disciplined AE services for design and construction phase services necessary to simultaneously execute task order requirements for a broad variety of projects that could include, but not limited to the following types of projects:

- Repair and/or replacement, modification/reconfiguration, and upgrades to NIST facilities (e.g., laboratories/scientific research, office, industrial). Including but not limited to mechanical, controls, electrical, plumbing, communications, fire alarm, and fire protection.
 - a. Incidental architectural, including historical assessments for preservation/rehabilitation/restoration of historic properties, space planning, interior design, civil, structural, and building enclosure (e.g., roofs, windows, foundations) work may be included.
- Repair and/or replacement, modification/reconfiguration for campus utility systems and supporting underground structures.
 - a. Incidental architectural, civil, structural work may be included.
- Program of Requirements (POR) development for Design/Build projects. Including meeting with stakeholders and confirming design intent for major systems (e.g., mechanical, electrical, plumbing, structural, controls, security, fire alarm, etc.)

Design Phase Services include engineering and engineering-related services such as:

- Surveying and investigations. Including hazardous material (e.g. asbestos, mold, lead paint) survey and risk assessment and topographical, boundary, or geotechnical surveying.
- Preparation of studies, reports, basis of design documents, drawings, specifications, cost estimates, calculations, and construction schedules
- Attending routine and/or issue-specific design review meetings. Record meeting minutes and distribute minutes to attendees.
- Evaluation and feasibility studies to include engineering, modeling and simulation
- Life cycle cost analysis identifying cost elements from initial concept through operations, support, and disposal
- Value engineering analysis to achieve design goals while lowering life cycle costs and maintaining the required performance, quality, reliability, and maintainability.

- Preparation of permit applications for Government submission to regulatory authorities based on specific project requirements.
- Additional engineering services as identified for a specific task order

Construction Phase Services include services related to a specific construction project such as:

- Material submittal reviews
- Request for Information (RFI) and Change Order request response
- Construction schedule analysis
- Attending routine and/or issue-specific construction meetings
- Construction quality assurance including on-site inspection of construction
- Record drawing preparation based on received construction as-builts
- Review of Commissioning plans prepared by third party and participation as part of the Commissioning Team (typical approach)
- Acting as the Commissioning Authority including development of commissioning specifications and initial commissioning plan. This role is anticipated to only be an occasional need that will be identified at the task-order level.
- Additional services as identified for a specific task order

The following are examples of potential tasks for this contract. Typical construction value for projects range between \$75,000 and \$5,000,000 but are not limited to this range. This list is intended to provide an idea of the type of work envisioned, but NIST makes no assurances that these projects will be included in this IDIQ.

1.1- Replace Potable Water Lines in Service Entry Rooms

2.1- Major Electrical Equipment Replacement

3.1- Short Circuit Study

4.1- Modify Electrical Relaying Study

5.1- Building Fire Alarm Panel and Audio/Visual Device Replacement

6.1- Laboratory and Office Renovations to meet Occupant's Needs

7.1- Air Handling Unit Replacement

8.1- Elevator Modernization

9.1- Replacement of Building Steam Stations and associated components

10.1-Restroom Renovations

11.1-Replace/Upgrade building-wide hydronic cooling systems

12.1- Redesign Mechanical Space to support replacement of failing HVAC unit

13.1- Replace building steam station and associated flash tanks

14.1-Installation of building-wide metering systems

VI. KEY PERSONNEL:

The Contractor shall employ the services of personnel submitted with the Standard Form 330, Section E and agreed upon during negotiations to perform the services required under this contract. All key personnel on the Contract shall be distinguished and licensed in their field (For non-architects and non-engineers, key personnel shall be credentialed in their field). Key personnel for this Contract shall be, at a minimum:

- A. Project Manager,
- G. Senior Mechanical (HVAC) Engineer,
- H. Senior Electrical Engineer,
- I. Senior Plumbing/Piping Engineer

No substitution will be made without the advance written approval of the Contracting Officer. Before approval, the Contractor must submit, in writing, the proposed replacement's experience and qualifications record.

For program of requirements individual task orders, the key personnel shall be required to have a minimum of Design - Build Institute of America (DBIA) Associate level certification.

VII. GENERAL REQUIREMENTS:

- A. AE Services**- The AE Contractor shall perform all or part of the following services as identified in each Task Order. The services described or specified shall not be deemed to constitute a comprehensive list having the effect of excluding services not specifically mentioned.

B. Project Management and Quality Services

- (1) The AE Contractor shall appoint a project coordinator or manager and alternate to serve as a single point of contact (POC) and liaison between the AE and the Contracting Officer (CO) and/or their representative for the services required under the Contract.
 - a. The project manager shall be responsible for the coordination of the services provided under the Contract and shall respond to questions regarding clarification of project deliverables during controls and review procedures to minimize conflicting errors and omissions and provide for the technical adequacy of design information.
- (2) Point of contact: The AE Contractor shall provide and designate a single point of contact (POC) to represent the AE's design team throughout the design and construction administration. The POC shall coordinate all communications between NIST and the AE and act as a central path of communication.
 - a. The designated POC shall communicate directly to the NIST through written requests and responses and provide documentation as requested to NIST in the form of a design progress/issues log. Upon request, an example of the approved issue log format will be provided.
 - b. The design progress/issues log shall:

- i. Summarize current design progress made during that week
 - ii. Log all sent and received requests for design information as well as other pertinent matters of communication
 - iii. Document all pending matters that will potentially affect the delivery schedule defined in the task document
 - iv. Identify upcoming actions that may require advanced preparations (i.e. on-site project meetings, site visits, and construction administration visits).
- (3) Meetings: Progress and review meetings shall be held at a mutually agreed upon time and location for discussion of questions and problems relating to the work under the Contract.
 - a. Attendees shall be key players for the task order in question; if the assigned expert is not available then an expert with equal qualifications may be substituted.
 - b. If the assigned design team/AE service team-member is not available, then an expert with equal qualifications may be substituted.
 - c. Attendees from the AE firm shall be conversant with the project and be of appropriate experience and skillset to the items discussed. Senior level AE staff that are knowledgeable about the project and specific task order shall be present for all meetings involving project decisions.
 - d. The AE shall take meeting notes for all meetings and provide a written copy within 5 business days to NIST.
- (4) Monthly Updates: The AE Contractor shall submit a monthly update report in writing for each individual task order. The report may be submitted by email to the Contracting Officer (CO), Contract Specialist (CS), and respective Contracting Officer Representative (COR) and shall address the following per project/task order:
 - a. Major Accomplishments, Significant Problems and Proposed Solutions, Issues with Contractual Requirements, Issues with Meeting the Task Order Deliverable Dates/Delays, and Contractor's Proposed Action to Remediate the Delays, if applicable.
- (5) Quality Assurance - The AE Contractor designated POC shall document and track all quality control feedback from NIST and coordinate responses to these items.
 - a. Quality control and assurance reporting shall be coordinated through the AE POC.
- (6) Coordination- When directed by the CO, the AE Contractor shall be required to coordinate with other consultants involved with specific design aspects of projects being developed under any separate Contracts.
- (7) Value Engineering- The AE Contractor shall cooperate with an "outside" (i.e., a third party) VE consultant when directed by the COR. This may require the A- to respond to VE recommendations and incorporated recommendations of the VE consultants into the design.
 - a. If the AE is required to incorporate VE changes to the design documents, this work may be included by a modification to the task order if so, determined by the CO.

C. Design Services

- (1) All designs, drawings, and specifications shall be prepared by the AE Contractor in accordance with applicable codes and regulations applicable to the state in which the project will be developed and shall conform to sound engineering practices.
 - a. All drawings shall be prepared using AUTODESK AUTOCAD 2025 (or latest version adopted by NIST) in accordance with the latest version of the U.S. Army Corps of Engineers Standard.
 - b. Specifications shall be prepared following the Construction Specifications Institute (CSI) Guidelines and in accordance with the latest version of MasterFormat or approved equivalent.
 - c. All criteria, factors, assumptions and codes upon which the design is based are to be documented and provided to NIST as well as the justification for the selection of these criteria.

D. Post-Construction-Award Services

- (1) The AE Contractor shall attend any Pre-Bid site visits as applicable and respond with recommendations for any questionable items. If necessary, develop amendments or assist with project procurement. A written response is required.
 - a. Attend as applicable pre-construction meetings, after award of construction Contract, and once prior to start of any additional phase.
 - b. Attend one (1) commissioning/project schedule coordination meeting after award of construction Contract (telephone attendance mandatory, site attendance optional).
- (2) Submittal, Request for Information (RFI), and Change Order Review (cost and no cost change orders)
 - a. The AE firm shall review and respond to all: submittals, change orders, commissioning reports, equipment startup reports, functional test reports, and commissioning issue reports.
 - b. All submittals, RFI's and change order proposals will be reviewed for compliance with the specifications and reasonable costs and returned to COR within an agreed timeframe after receipt.
 - c. The AE shall support commissioning inquiries related to the designed mechanical, electrical, plumbing, fire protection, lighting, and other building systems included in this project's scope. Where required by the commissioning authority, the A- shall actively respond to design intent clarification requests that occur while inspecting and testing any systems.
- (3) Commissioning Support- As applicable, the AE Contractor shall work with the commissioning agent throughout the design and construction process to ensure all sustainable objectives are met. The AE shall support construction phase commissioning project meetings, reports and inquiries.

- (4) Construction Quality Management (CQM)- The AE Contractor shall provide CQM services as needed, not limited to the following tasks:
- a. Scheduling Support - regularly review, analyze, and monitor performance schedules, including, but not limited to the preliminary/baseline schedule, Critical Path Method (CPM) construction schedules, and fragnets as required.
 - b. Quality Construction Management, Reviews and Administration Assistance - shall include inspections, verifications, details audits and evaluations of materials and workmanship necessary to determine and document the quality of construction. The CQM shall use its expertise and knowledge of construction practices, means and methods to execute effective Quality Assurance.
- (5) Site Visits (In addition to site visits listed in this section)
- a. The AE Contractor shall visit the site at the request of the COR or CO to answer technical questions and/or provide clarification to documents due to errors or omissions at no additional cost to the government.
 - b. Attend and assist with site inspections as requested by the COR or CO at the interim completion levels and final inspection of percent complete for all construction projects.

All A/E services provided under this requirement shall be consistent with applicable policies, guidelines and/or procedures for the design and construction of NIST facilities and infrastructure as well as all applicable Federal Regulations. Any applicable guidelines, policies, and or procedures will be provided at the Order level.

Government Responsibilities

a. Administration of Contract

The Contracting Officer is responsible for the administration of the Contract. No change shall be made in the provision of this Contract without written authorization of the Contracting Officer.

- ii. Re-submittal – In the event that documents submitted for review, as described in this document, are deficient or incomplete as specifically defined in the task order for a particular stage of completion, the AE shall correct the deficiencies and resubmit the documents. No partial re-submittals will be permitted. The Contractor shall assume all costs and schedule requirement due to the re-submittal of documents.

b. Government Furnished Information

The Government, through the office of the Contracting Officer, will furnish within five days after issuance of a Task Order, applicable data and information (as available) for the execution of the work described. Data shall be returned to the Contracting Officer upon completion of the individual work order.

- iii. Data will include where applicable and available the following:
 - a. General Site Plan
 - b. Small scale layout drawing or sketch showing possible arrangement of space (to serve as guide only).
 - c. Basic floor plan showing locations of project (as required).
 - d. Existing record drawings as available.
 - e. Topographic data as available.
 - f. Existing utilities information as available.
 - g. Special requirements for preparation of drawings, reports, specifications, cost estimates, and design analysis.
 - h. Government-furnished equipment list, if applicable.
- i. Accuracy of Planning Information. The AE shall visit the project site to investigate the information shown on the Government Furnished drawings, record (as- built) drawings and other planning documents which are part of this Contract. This information is the best available, but the Government does not guarantee its accuracy or completeness. The AE investigation of field conditions shall be performed in a competent professional manner in accordance with the Guidelines for OFPM Contractor and A/E Firms Performing Site Investigations, Fieldwork, and Construction at the National Institute of Standards and Technology.
- ii. Discrepancies in Planning Information. The AE shall promptly report to the Contracting Officer in writing any discrepancies between this Contract and the planning information provided by the Government. The AE shall make no adjustments for this work due to the discrepancy before the Contracting Officer has reviewed the matter and forwarded his determination to the AE. The AE's failure to report any such discrepancy discovered or to wait for Contracting Officer's determination shall be at his risk and expense.
- iii. The AE shall always maintain close liaison with the COR by the appropriate means to assure maximum coordination in accomplishment of the work.
- iv. The AE shall advise the COR of any discrepancies, ambiguities and lack of clarity noted in plans and specifications furnished for use in connection with each Task Order.
- v. During the execution of the work under each Task Order, the AE shall keep in close liaison with the COR who will coordinate the work with the using agency and other agencies.
- vi. All letters, reports, plans, sketches, specifications, estimates,

computation sheets and other documents prepared by the A-E shall be marked "FOR OFFICIAL USE ONLY".

Any other Government furnished property, material, equipment, or information needed in the performance of this requirement will be detailed in each task order.

VIII. SCHEDULE OF DELIVERABLES:

Specific schedule of deliverables shall be defined for each task order.

Standards of Acceptance: The NIST POC or COR shall review all the above deliverables and respond with an acceptance or request for revision email to the Contractor Point of Contact (POC) within 21 days of receipt of deliverable unless otherwise defined at the task order level. This section applies to deliverables which must be reviewed and approved for acceptance (e.g., draft plans, drawings, etc.).

IX. DELIVERY TERMS FOR ITEMS/PRODUCTS:

Delivery shall be F.O.B Destination and shall occur in accordance with the delivery due dates provided at the task order level.

FOB Destination means: The AE shall pack and mark the shipment in conformance with carrier requirements, deliver the shipment in good order and condition to the point of delivery specified in the purchase order, be responsible for any loss of and/or damage to the goods occurring before receipt and acceptance of the shipment by the consignee at the delivery point specified in the purchase order; and pay all charges to the specified point of delivery.

The AE shall deliver all Line Items to:
The National Institute of Standards and Technology
100 Bureau Drive, Building 301
(C/O POC/COR Name and Building Number)
Gaithersburg, MD 20899-1640

*****Due to ongoing limited site access of the NIST facilities, the Awardee SHALL confirm with the Designated Government Official and the Contracting Officer prior to ANY shipment to NIST*****

X. SPECIAL REQUIREMENTS:

a. Architect-Engineer Qualifications

The AE must submit a completed Standard Form 330, which will contain information about their professional qualifications, to be considered for award as part of the IDIQ Evaluation Process.

b. Ownership of Documents and Intellectual Properties

NIST has ownership of the rights to all CAD files, BIM Model, engineering calculations and Facility Data developed for the Project. NIST may make use of this data following any deliverables. All designs, drawings, specifications,

notes, and other work developed in the performance of this Contract shall be and remain the sole property of the Government and may be used on any other work without additional compensation. With respect thereto, the AE agrees not to assert any rights and not to establish any claim under the design patent or copyright laws. The AE, for a period of 3 years after project closeout, agrees to furnish and provide access to all retained materials at the request of the Contracting Officer. Unless otherwise provided in this Contract, the AE shall have the right to retain copies all such materials beyond such period.

c. Task Order Procedures

The following is a preliminary guide of how procedures will be followed for issuance of a Task Order under this Contract unless otherwise notated at the Task Order level:

1. The Contracting Officer will issue a "Request for Task Order Proposal." A statement of work and any other related project documents will be attached to the notice. The AE shall review the documents attached to the notice and formulate any questions that they may have regarding the project.
2. The AE shall identify any issues with the Statement of Work and review the contracting procedures for the project. The AE will have a chance to perform some preliminary Site Investigation necessary to provide a cost proposal.
3. AE will submit a technical proposal based upon the evaluation criteria provided with the task order documents.
4. Government will review the technical proposal to select the most-qualified firm.
5. The most-qualified firm will provide a cost proposal.
6. Government will review the cost proposal and schedule negotiations as necessary.
7. Government will issue a Task Order

d. Liability

The AE shall be liable for, and shall indemnify and hold harmless the Government against, all actions or claims for loss of or damage to property or the injury or death of persons, arising out of or resulting from the fault, negligence, or act or omission of the A-E, its agents, or employees.

XI. SAFETY ON THE NIST CAMPUS:

NIST is committed to maintaining a work environment that safeguards the public and the environment, as well as personnel and property. The AE employee must be responsible for knowing and complying with all installation safety prevention regulations. Such regulations include, but are not limited to, general safety, fire prevention, and waste disposal.

If the task order requires field investigation by the Contractor, including subcontractors, at NIST-owned and operated sites, the Contractor shall:

Complete the “A/E Design Consultant Safety Checklist” and provide Activity Hazard Analysis (AHA) if required by the checklist based on the planned activities. Provide this information within 7 days of task order award.

If required for field investigations and/or site visits, the AE shall provide records of applicable training for their personnel (including subcontractors). This may include training for proper use of PPE, respirators, fall protection, electrical safety, or other safety-related topics. AE personnel (including subcontractors) may be required to attend NIST-specific health or safety training that is identified at the task-order level.

NIST will provide a copy of the signed checklist or will provide written comments regarding the AHA within seven (7) days of receipt. The AE will be required to submit a revised AHA within seven (7) days after that to NIST for their review. This cycle may be repeated, if necessary, until NIST determines that the safety plan to be acceptable.

Safety-Related Equipment

The AE Contractor shall provide all safety-related equipment required to perform the work under this award, including personal protective equipment (PPE) and other equipment that is required or appropriate to protect contractor personnel and other personnel on the NIST campus.

Contractor-provided personal protective equipment may include hard hats, safety shoes, safety glasses (prescription or standard), respiratory protection (powered air purifying respirator or full-face mask respirator), high visibility clothing and gloves that meet applicable OSHA standards for PPE.

Health Hazard Assessments and Medical Evaluations

If PPE is required to protect personnel from the environment being entered, the AE shall provide required health hazard exposure assessments, medical evaluations and screenings, and exposure monitoring required by applicable OSHA requirements. Ensure PII information is protected.

XII. SECURITY:

NIST is a restricted campus. An identification badge is required for access and entry into buildings and is shown to the armed Security Police when entering the campus. The AE (and/or any subcontractor) and its employees who require physical access to NIST facilities or logical access to NIST IT networks or systems must comply with Homeland Security Presidential Directive (HSPD)-12, *Policy for a Common Identification Standard for Federal Employees and Contractors*; OMB M-05-24; OMB M-19-17; FIPS 201, *Personal Identity Verification (PIV) of Federal Employees and Contractors*; NIST HSPD-12 policy; and *Executive Order 13467, Part 1 §1.2*. Reference the “Guidelines for OFPM Contractors and A/E Firms Performing Site Investigations, Fieldwork, and Construction at the National Institute of Standards and Technology” for further information about badging requirements.

All AE (and/or any subcontractor) employees must obtain a background investigation commensurate with their position sensitivity designation that complies with Parts 1400 and 731 of Title 5, Code of Federal Regulations (CFR). The sensitivity designation levels will be determined at the task-order level.

All AE employees must register their vehicles with the NIST Security Office to gain access to the campus. A valid driver's license, Government-furnished civilian ID, proof of insurance and current registration must be presented to the NIST Security Office, at which time a NIST vehicle pass will be issued. The pass must be displayed on the vehicle in accordance with NIST Security Office instructions.

The AE (and/or any subcontractor) must submit a roster by name, position, e-mail address, phone number and responsibility, of all staff working under this acquisition where the Contractor will develop, can access, or host and/or maintain a government information system(s).

The roster must be submitted to the COR and/or CO within 14 days of the effective date of contract award. Any revisions to the roster as a result of staffing changes must be submitted within 14 days of the change. The COR will notify the Contractor of the appropriate level of investigation required for each staff member. If the employee is filling a new position, the Contractor must provide a position description, and the Government will determine the appropriate suitability level.

XIII. PAYMENT SCHEDULE:

The Government shall pay the AE the contract price per the appropriate clauses as provided in the Contract. Invoice instructions will be provided for each Task Order.

Advance payment is not authorized unless otherwise approved. The AE must invoice in the arrears according to the payment schedule. The AE shall be paid, in accordance with Net 30-day payment terms, upon receipt and acceptance of a proper invoice, in accordance with the following schedule:

1. Submission of each deliverable.
2. If applicable, successful demonstration of any technical requirements set forth at the task order level.
3. After receiving an invoice submitted properly, according to all invoice terms and conditions.

NOTE: Partial shipments and partial invoices will not be accepted, unless otherwise requested and accepted by the Contracting Officer prior to award offer. Proposed payment schedules shall be submitted with vendor's response to the RFQ for consideration.

Federal Holidays

The NIST campus will be closed on all federal holidays. When a holiday falls on a weekend, the holiday is observed on either the Friday prior or Monday after the holiday. Below is a list of federal holidays:

1. **New Year's Day**
2. **Birthday of Martin Luther King, Jr.**
3. **Inauguration Day – occurs every fourth year on January 20th**
4. **Washington's Birthday**
5. **Memorial Day**
6. **Juneteenth National Independence Day**
7. **Independence Day**
8. **Labor Day**
9. **Columbus Day**
10. **Veterans Day**
11. **Thanksgiving Day**
12. **Christmas Day**

End of Statement of Work